

(3.1)

(b) $\Rightarrow$ (a)

$$B_1 = (\vec{u}_1, \dots, \vec{u}_n)$$

$$B_2 = (\vec{v}_1, \dots, \vec{v}_n) \quad \text{jsoú báze}$$

Nechť $G_1 = G_2$,

$$\text{tj. } \langle \vec{u}_j, \vec{u}_k \rangle = \langle \vec{v}_j, \vec{v}_k \rangle \quad \forall j, k \in \{1, \dots, n\}$$

$$\text{Nechť } f: \mathbb{R}^n \rightarrow \mathbb{R}^n: \vec{u}_i \mapsto \vec{v}_i$$

$$f(\vec{u}_i) = \vec{v}_i \Rightarrow Q\vec{u}_i = \vec{v}_i$$

↑
lineární zobrazení je lineární zobrazení
a existuje.
Jelikož matice Q je

$$\begin{aligned} [f]_{B_1}^{B_2} &= [id]_{B_2}^{B_2} [f]_{B_2}^{B_1} [id]_{B_1}^{B_1} = [id]_{B_2}^{B_2} I_n [id]_{B_1}^{B_1} = [id]_{B_2}^{B_2} \cdot [id]_{B_1}^{B_1} \\ &\stackrel{||}{=} Q \end{aligned}$$

zřejmě ani důležitá,

že tato je OG matice

(její sloupce tvoří ON posklápanost)

$$([\vec{v}_1]_{B_1} | [\vec{v}_2]_{B_1} | \dots | [\vec{v}_n]_{B_1}) \cdot ([\vec{e}_1]_{B_2} | [\vec{e}_2]_{B_2} | \dots)$$

$$(\vec{v}_1 | \vec{v}_2 | \dots | \vec{v}_n) \cdot ([\vec{e}_1]_{B_1} | [\vec{e}_2]_{B_1} | \dots)$$

$$Q^*Q = ([id]_{B_2}^{B_2} \cdot [id]_{B_1}^{B_1})^* ([id]_{B_2}^{B_2} \cdot [id]_{B_1}^{B_1}) =$$

$$= [id]_{B_1}^{B_1^T} \cdot [id]_{B_2}^{B_2^T} \cdot [id]_{B_2}^{B_2} \cdot [id]_{B_1}^{B_1} =$$

$$G_2 = G_1 = [id]_{B_1}^{B_1^T} \cdot [id]_{B_1}^{B_1}$$

$$= ([id]_{B_1}^{B_1} [id]_{B_1}^{B_1})^T I_n = I_n^T I_n = I_n$$

Tvzení 8.82 (2)

 $\implies Q$ je ortogonální.

$$(a) \exists Q_{n \times n} \text{ l.i. } Q \vec{u}_i = \vec{v}_i \text{ pro } \forall i \in \{1, 2, \dots, n\} \Rightarrow \vec{u}_i = Q^* \vec{v}_i \quad \text{Thm 8.82 (2)}$$

$$(b) G_1 = G_2$$

$$(a) \Rightarrow (b): (G_1)_{j,k} = \langle \vec{u}_j, \vec{u}_k \rangle = \langle Q^* \vec{v}_j, Q^* \vec{v}_k \rangle = \langle \vec{v}_j, \vec{v}_k \rangle = (G_2)_{j,k} \quad \uparrow \quad \text{Thm 8.82 (6)}$$